

Comptia A+

Assignment A+ - Understanding and maintenance of Networks

1. List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

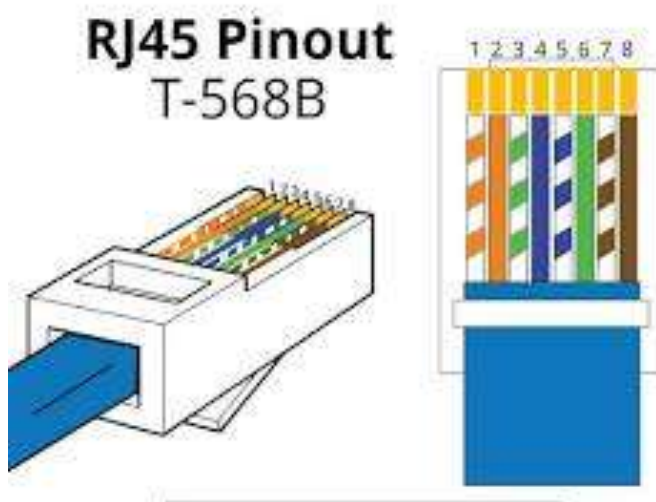
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Feature	Ethernet Cable	Fiber Optic Cable
1. Transmission	Uses electrical signals to transmit data.	Uses light signals to transmit data.
2. Speed & Distance	Suitable for shorter distances and common network connections.	Supports very high speeds and much longer distances.
3. Cost & Installation	Generally cheaper and easier to install.	Generally more expensive and requires specialized equipment.

2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.

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1. T568B Ethernet Cable Wiring Diagram



2. Pin-Out Configuration Table (T568B Standard)

Both ends of a straight-through cable follow the exact same pin layout:

Pin Number	Wire Color Code	Signal Function
Pin 1	White / Orange	Transmit Data (+)
Pin 2	Solid Orange	Transmit Data (-)
Pin 3	White / Green	Receive Data (+)
Pin 4	Solid Blue	Bidirectional Data
Pin 5	White / Blue	Bidirectional Data
Pin 6	Solid Green	Receive Data (-)
Pin 7	White / Brown	Bidirectional Data
Pin 8	Solid Brown	Bidirectional Data

3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

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Video Reference:

Title: RJ45 Crimping Tutorial (Ethernet Cable Making)

URL: <https://www.youtube.com/watch?v=9hpPczmnDUE>

Tools Required:

- CAT6 Ethernet cable
- RJ45 connector
- RJ45 crimping tool
- Cable stripper/cutter

- Cable tester (optional)

Procedure:

1. Cut the CAT6 cable:

Cut the CAT6 cable to the required length using a cable cutter.

2. Strip the outer jacket:

Use the stripper on the crimping tool to carefully remove approximately 2–3 cm of the outer jacket. Be careful not to damage the internal wires.

3. Separate the wires:

Untwist the four pairs and gently straighten the eight individual wires.

4. Arrange the wires according to T568B:

Arrange the wires from left to right in this order:

White/Orange → Orange → White/Green → Blue → White/Blue → Green → White/Brown → Brown.

5. Trim the wires:

Hold all eight wires together and cut them evenly so that they are the same length.

6. Insert the wires into the RJ45 connector:

Carefully insert all eight wires into the RJ45 connector. Make sure each wire remains in the correct color order and reaches the end of the connector.

7. Check the connection:

Before crimping, check that the wires are correctly positioned and that the cable jacket enters the connector far enough for the strain relief to grip it.

8. Crimp the connector:

Place the RJ45 connector into the correct slot of the crimping tool and firmly squeeze the handles until the connector is securely crimped.

9. Test the cable:

If a cable tester is available, connect both ends of the cable to the tester and check that all eight connections are detected correctly.

4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences. Consider speed, distance, and budget for your answer.

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Selected Cable Type: Ethernet (Cat6 / Cat6a)

Selected Tool: RJ45 Crimper

Explanation:

For a 10-PC gaming café, copper Ethernet cabling combined with an RJ45 crimper is the ideal choice because it easily handles Gigabit speeds over short indoor distances at a fraction of the cost of fiber optics. Fiber optic cables and specialized fiber cleavers are designed for long-distance backbones and multi-gigabit campus networks, making them an unnecessarily expensive and complex setup for a single-room LAN. Using standard Cat6 Ethernet cables allows for fast 10 Gbps speeds, low latency for gaming, and quick, budget-friendly installation and maintenance using a standard RJ45 crimper.